Aix-Marseille School of Economics

Methodology in Econometric and Statistical Studies

Analysis and Prediction of Unemployment Dynamics

A Combined Time-Series and Cross-Sectional Approach

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Course:

M2 Econometrics

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Academic Year: 2025–2026

January 30th, 2026

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1 Introduction

Understanding and predicting unemployment dynamics is a central issue in applied economics, as unemployment reflects both the state of the business cycle and the structural functioning of the labor market. Accurate unemployment forecasts are crucial for policymakers, central banks, and public institutions, as they guide decisions related to monetary policy, labor market interventions, and social protection systems. Moreover, unemployment is influenced by both macroeconomic conditions and individual characteristics, making it a natural candidate for complementary time-series and cross-sectional analyses.

The objective of this project is to analyze and predict unemployment using two complementary empirical approaches. First, a time series analysis is conducted to model and forecast aggregate unemployment dynamics over time using macroeconomic indicators. Second, a cross-sectional analysis is performed to study individual-level determinants of unemployment, focusing on demographic and socioeconomic characteristics. Combining these two perspectives allows for a richer understanding of unemployment, capturing both its aggregate evolution and its microeconomic foundations.

The time series analysis relies on monthly macroeconomic data obtained from the OECD database, covering a long historical period. The main variable of interest is the total unemployment rate, observed at a monthly frequency. To explain and predict unemployment, the analysis incorporates lagged values of unemployment itself, capturing persistence and inertia, as well as price indicators such as the Consumer Price Index (CPI), which reflect inflationary conditions and the stance of monetary policy. This approach is motivated by standard macroeconomic theory and empirical evidence, according to which unemployment exhibits strong temporal dependence and may respond indirectly to inflation through economic activity and policy channels. Time series models such as autoregressive (AR) and autoregressive models with exogenous variables (ARX) are particularly well suited for this purpose.

The cross-sectional analysis is based on individual-level survey data describing labor market outcomes and personal characteristics, including age, sex, race, education, and employment status. This dataset allows us to construct a binary unemployment indicator at the individual level and to analyze how the probability of being unemployed varies across population groups. Cross-sectional econometric models are used to identify key socioeconomic determinants of unemployment and to quantify disparities across demographic categories. This micro-level approach complements the time series analysis by providing insights into the structural and distributional aspects of unemployment that are not visible in aggregate data.

The combination of these two approaches is both methodologically coherent and economically relevant. While time series models focus on predicting aggregate unemployment and capturing macroeconomic dynamics, cross-sectional models explain individual unemployment risk and heterogeneity within the labor market. Together, they provide a comprehensive framework for understanding unemployment from both a macroeconomic and a microeconomic perspective. The remainder of the report is organized as follows. Part I presents the time series analysis: data description, model specification, estimation, forecasting, and model comparison. Part II focuses on the cross-sectional analysis, describing the individual-level data, the construction of unemployment indicators, the econometric models, and the main empirical findings.

The final section concludes and discusses policy implications derived from both approaches.

2 Literature Review

The study of unemployment is based on the analysis of its temporal persistence and its interactions with other economic aggregates. This literature review examines the theoretical and empirical foundations that justify the approach adopted in this report.

Macroeconomic Determinants and Causality (Time Series)

The article by Sahoo and Sahoo (2019), *"The relationship between unemployment and some macroeconomic variables: Empirical evidence from India"*, is particularly relevant to our time-series work. The authors analyze the impact of variables such as real GDP (RGDP), inflation (CPI), and gross fixed capital formation (GFCF) on unemployment for the period 1991–2017.

Their results, obtained through Augmented Dickey-Fuller (ADF) stationarity tests, show that all variables become stationary after first differencing. This validates our graphical approach in Figure 5 to stabilize variance. Furthermore, the Granger causality test reveals a unidirectional relationship from RGDP to unemployment, confirming economic growth as an essential predictive driver of the labor market. The authors also note a bidirectional causality between the labor force (LF) and unemployment, reinforcing the idea that these variables influence each other]. Their use of a VECM model shows an intercept β_0 of 0.147, indicating proportionately increasing unemployment during the study period, which justifies our choice of AR(1) and ARX models.

Microeconomic Determinants and Individual Heterogeneity (Cross-Section)

To complement the macroeconomic view, the article by Ousmane Marikol (2023), *"Unemployment in the WAEMU Countries: A Cross-Sectional Data Approach"*, provides an empirical evaluation of microeconomic determinants. Using multinomial logistic analysis, the study finds that being female, single, young, disabled, or living in urban areas significantly increases the risk of unemployment and inactivity.

Notably, the study observes that higher education graduates in certain regions are 4.2 times more likely to be unemployed than non-graduates, often due to an inability of education systems to reconcile training with business needs. Marital status also plays a role, with married workers being 0.3 times less likely to be unemployed, possibly due to family responsibilities making them less demanding regarding job quality. This micro-level evidence links directly to our cross-sectional analysis of age, sex, and education.

In summary, these works confirm that:

- **Macro-level**: Inflation (CPI) and investment (GFCF) have significant long-run relationships with unemployment, justifying their use in our ARX and forecasting models[.
- **Micro-level**: Individual characteristics like gender, education, and age explain disparities in unemployment risk not visible in aggregate data.
- **Education**: While education reduces long-term unemployment risk globally, a mismatch between skills and market needs can paradoxically increase graduate unemployment in specific contexts.

3 Time Series: Macroeconomic Unemployment

3.1 Data Description and Motivations

3.1.1 Data description and sources

The time series analysis relies on monthly macroeconomic data obtained from the OECD database. The dataset covers a long historical period starting in January 1955 and provides consistent monthly observations on labor market outcomes and prices. The monthly frequency is particularly well-suited for short-term monitoring and forecasting of unemployment, as it allows for the identification of cyclical movements and turning points in the labor market.

The database contains the following variables:

- A time identifier (Time period), reported in year–month format.
- The unemployment rate for women.
- The unemployment rate for men.
- The total unemployment rate.
- The Consumer Price Index (CPI), with a base year equal to 2015.

All unemployment variables are expressed as percentages of the labor force, while the CPI is an index capturing the evolution of consumer prices over time.

3.1.2 Variable of interest

The main variable of interest in the time series analysis is the total unemployment rate. This variable is chosen as the dependent variable for several reasons. First, it provides a comprehensive measure of labor market conditions at the aggregate level, summarizing the overall state of employment in the economy. Second, it is the key indicator typically monitored by policymakers, central banks, and international organizations when assessing economic activity and designing macroeconomic policies. Finally, the total unemployment rate exhibits strong persistence and cyclical behavior, making it particularly suitable for time series modeling and forecasting.

Although the dataset also includes unemployment rates by gender, the focus of the baseline time series models is on aggregate unemployment in order to keep the specification parsimonious and directly comparable to standard macroeconomic forecasting exercises.

3.1.3 Predicting variables and model strategy

The set of predicting variables is constructed to capture both the dynamic persistence of unemployment and the influence of the macroeconomic environment.

First, lagged values of the total unemployment rate are included as explanatory variables. From an economic perspective, unemployment is known to slowly adjust over time due to labor market frictions, job matching processes, and institutional rigidities. From a statistical perspective, unemployment series display strong autocorrelation, which justifies the use of auto-regressive (AR) models. Including lags of the dependent variable allows the model to exploit the information contained in past unemployment realizations when predicting future outcomes.

Second, the Consumer Price Index (CPI) is introduced as an additional explanatory variable. The CPI reflects inflationary pressures and the overall price environment in the economy. Inflation may affect unemployment indirectly through monetary policy reactions and real economic activity.

For instance, periods of rising inflation are often associated with tighter monetary policy, which may dampen economic growth and increase unemployment. The CPI is therefore included with a lag, ensuring that only information available at time $t-1$ is used to predict unemployment at time t.

3.1.4 Temporal structure and relevance of the time dimension

The time dimension plays a central role in this analysis. All variables are observed at a monthly frequency, and the models explicitly exploit the chronological ordering of the data. Lagged variables are constructed to ensure a clear information structure, where predictions at time t are based exclusively on information available up to time $t-1$.

This temporal structure is essential for producing meaningful out-of-sample forecasts and avoiding look-ahead bias.

The long time span of the dataset allows the models to capture different economic regimes, including periods of high inflation, economic expansions, recessions, and major shocks. As a result, the time series framework provides a natural and coherent setting for studying the dynamics of unemployment and evaluating its predictability over time.

3.2 Model specification and assumptions

This section presents the econometric models used to analyze and predict the dynamics of unemployment, as well as the underlying assumptions required for estimation and inference.

3.2.1 Model specification

The starting point of the analysis is the observation that unemployment exhibits strong persistence over time. To capture this dynamic behavior, the baseline specification relies on autoregressive models, in which current unemployment depends on its own past realizations.

The first model is a simple autoregressive model of order one, denoted AR(1):

$$u_t = \beta_0 + \beta_1 u_{t-1} + \epsilon_t \tag{1}$$

where u_t represents the total unemployment rate at time t, β_0 is a constant term, and ϵ_t is an error term capturing unexpected shocks to unemployment.

To allow for richer dynamics, a second specification includes an additional lag of unemployment, leading to an autoregressive model of order two:

$$u_t = \beta_0 + \beta_1 u_{t-1} + \beta_2 u_{t-2} + \epsilon_t \tag{2}$$

Including multiple lags allows the model to capture more complex adjustment processes and delayed responses of the labor market.

In order to account for the macroeconomic environment, an extended specification incorporates an exogenous price indicator. This leads to an autoregressive model with an exogenous variable (ARX):

$$u_t = \beta_0 + \beta_1 u_{t-1} + \gamma CPI_{t-1} + \epsilon_t \tag{3}$$

where CPI_{t-1} denotes the lagged Consumer Price Index. The use of a lagged CPI ensures that the model only relies on information available prior to period t, which is essential for forecasting purposes. This specification allows inflation-related conditions to affect unemployment indirectly, for instance through monetary policy responses and economic activity.

All models are estimated using monthly data, and the lag structure is chosen to balance model parsimony and predictive performance.

3.2.2 Information structure and forecasting framework

The models are designed within a predictive framework where the information set available at time $t-1$, denoted I_{t-1}, includes past values of unemployment and the CPI. Predictions of unemployment at time t are therefore based exclusively on lagged variables. This ensures that the forecasting exercise is realistic and free from look-ahead bias.

3.2.3 Assumptions on the error term

The validity of the estimation and prediction results relies on standard assumptions regarding the error term ϵ_t.

First, the error term is assumed to satisfy a conditional mean restriction:

$$E[\epsilon_t|I_{t-1}] = 0 \tag{4}$$

This assumption implies that the regressors are exogenous with respect to the error term, meaning that all systematic information relevant for predicting unemployment is captured by the included variables.

Second, the error term is assumed to have constant conditional variance:

$$Var(\epsilon_t|I_{t-1}) = \sigma^2 \tag{5}$$

This homoscedasticity assumption simplifies inference but may be violated in practice, particularly during periods of economic turmoil or structural change.

Finally, for inference and the construction of prediction intervals, the error term is often assumed to be approximately normally distributed. While normality is not strictly required for consistency of the OLS estimator, it facilitates statistical inference and the interpretation of confidence and prediction intervals.

3.2.4 Estimation method and properties

All models are estimated using the Ordinary Least Squares (OLS) method. Under the assumptions of linearity, exogeneity, and homoscedasticity, the OLS estimator is given by:

$$\hat{\beta} = (X'X)^{-1}X'Y \tag{6}$$

and is the Best Linear Unbiased Estimator (BLUE) according to the Gauss–Markov theorem. OLS is particularly attractive in this context due to its simplicity, transparency, and widespread use in time series forecasting with autoregressive structures.

However, potential issues such as autocorrelation of residuals, non-stationarity of some variables, and structural breaks may affect the efficiency of the estimator and the reliability of inference. These limitations are discussed when interpreting the estimation and forecasting results.

3.3 Implementation, Code, and Interpretation of the Results

3.3.1 Implementation and estimation procedure

The empirical implementation is conducted using the statistical software R. The dataset is first organized as a monthly time series by converting the time identifier into a proper date format and sorting observations chronologically. Lagged variables of the unemployment rate and the CPI are then constructed to respect the information structure of the forecasting framework.

The estimation strategy follows four main steps:

1. **First**, descriptive analyses are performed to visualize the evolution of unemployment and inflation over time.

2. **Second**, stationarity properties of the main variables are assessed using Augmented Dickey–Fuller (ADF) tests.

3. **Third**, the sample is divided into a training period and a test period to evaluate out-of-sample predictive performance.

4. **Finally**, several autoregressive models are estimated using Ordinary Least Squares (OLS), and their predictions are compared using standard loss functions.

The models are estimated on the training sample, while predictions are generated for the test period to avoid overfitting and ensure a realistic forecasting exercise.

3.3.2 Descriptive analysis

The following analysis provides a visual overview of the key trends and statistical properties of the variables used in the time-series models.

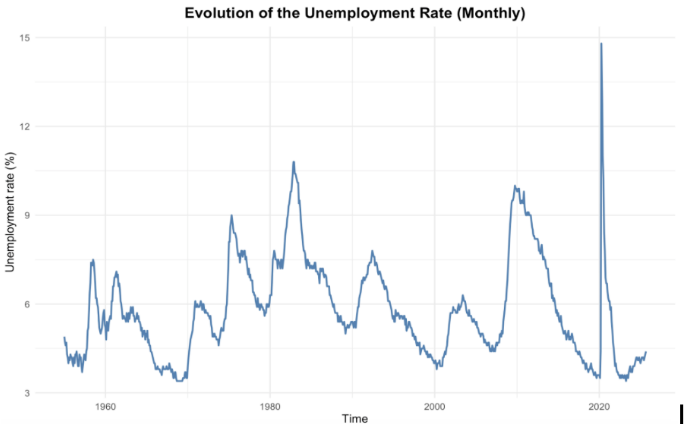

Figure 1 – Evolution of the unemployment rate (monthly) The figure displays the monthly evolution of the total unemployment rate over a long historical period. Several important features emerge from this graph.

First, the unemployment rate exhibits clear cyclical behavior, with recurrent phases of increase and decrease corresponding to economic expansions and recessions. Peaks in unemployment are observed during major economic downturns, while troughs correspond to periods of sustained economic growth.

Second, unemployment shows strong persistence over time. Increases in unemployment are typically followed by gradual declines rather than abrupt reversals, suggesting slow adjustment processes in the labor market. This persistence is consistent with the presence of labor market frictions, such as job matching costs and institutional rigidities.

Finally, the series is characterized by exceptional shocks, most notably a sharp spike around 2020, associated with the COVID-19 crisis. This episode highlights the presence of structural breaks and extreme events that may affect both estimation and forecasting performance. Overall, this figure suggests that unemployment dynamics are driven by inertia and cyclical forces, making time series models particularly appropriate for analysis and prediction.

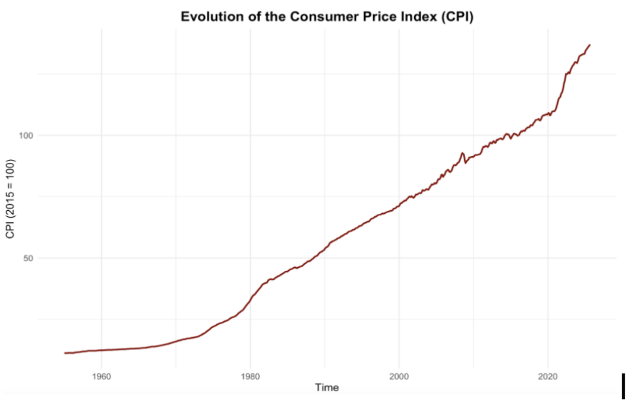

Figure 2 – Evolution of the Consumer Price Index (CPI) The figure displays the monthly evolution of the total unemployment rate over a long historical period. Several important features emerge from this graph.

This persistent increase reflects cumulative inflation over time. The absence of visible mean reversion and the presence of a clear trend indicate that the CPI is likely non-stationary in levels. Periods of steeper increases can be observed, notably during inflationary episodes and more recent years, while some stabilization appears during other periods. From an econometric perspective, this visual evidence suggests that the CPI may need to be treated with caution in regression models, either by using lagged values, growth rates, or by explicitly testing its stationary properties. This motivates the subsequent formal stationary analysis.

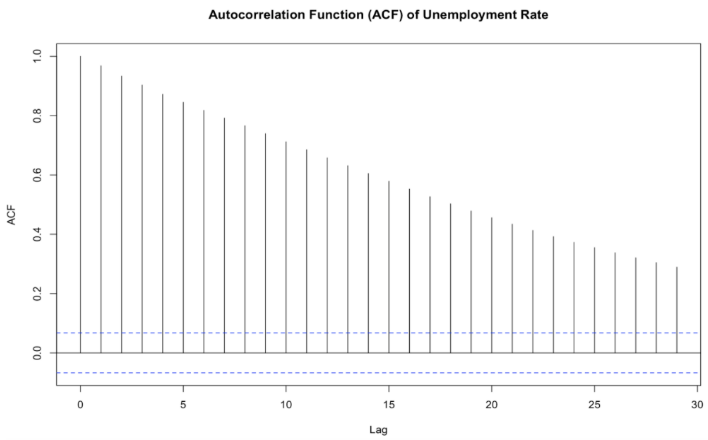

Figure 3 – Autocorrelation Function (ACF) of the unemployment rate The ACF plot shows very high autocorrelation at short lags, followed by a slow and gradual decay as the lag length increases.

Autocorrelation remain statistically significant for many lags. This pattern is characteristic of a highly persistent time series and strongly suggests that current unemployment depends heavily on its past values. The slow decay of the ACF is consistent with an auto regressive process rather than a white noise series. Such behavior provides strong empirical justification for the use of auto regressive (AR) models, as past unemployment rates clearly contain valuable information for predicting future unemployment

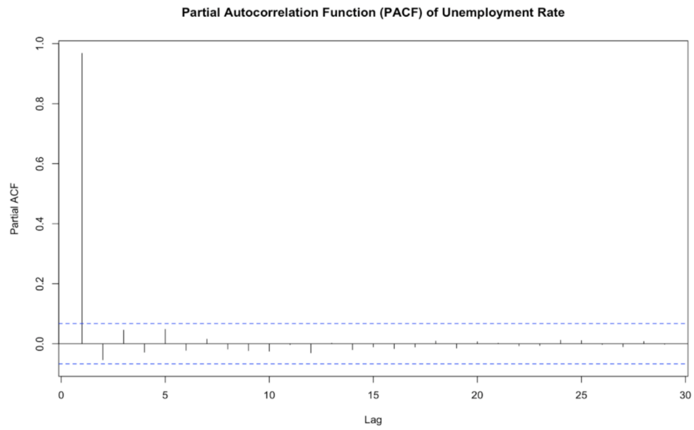

Figure 4 – Partial Autocorrelation Function (PACF) of the unemployment rate The PACF plot displays a large and significant spike at lag 1, while subsequent lags are close to zero and generally not statistically significant.

This pattern is typical of an AR(1) process, where the first lag captures most of the dependence structure in the data. The absence of strong partial autocorrelation at higher lags suggests that adding many lags may not substantially improve the model. From a modeling perspective, this result supports the use of parsimonious specifications, such as AR(1), while still allowing AR(2) models to be tested as robustness checks.

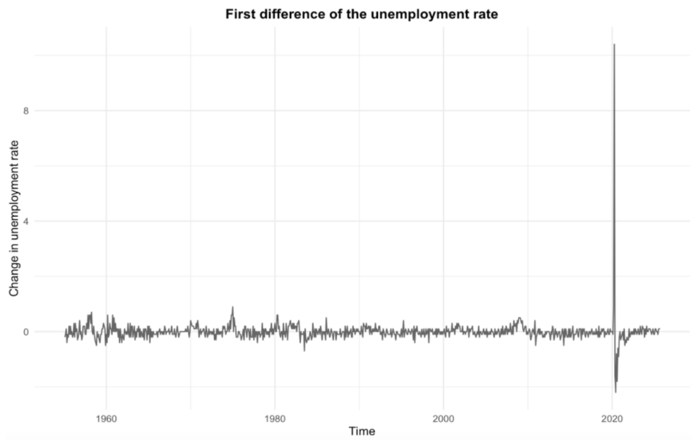

Figure 5 – First difference of the unemployment rate The final figure shows the first difference in the unemployment rate. The different series fluctuates around zero, without an obvious trend, indicating that differencing substantially reduces persistence. Most variations are relatively small, except during exceptional periods, such as the COVID-19 crisis, where abrupt changes occur. This suggests that while the level of unemployment is highly persistent, changes in unemployment are more stable over time.

The contrast between the level series and the different series provides further evidence of strong persistence in unemployment and raises the question of whether the unemployment rate should be treated as a stationary or near-unit root.

Together, the descriptive analysis highlights several key features of the data. The unemployment rate exhibits strong persistence and cyclical behavior, as confirmed by the time series plot and the ACF and PACF functions. In contrast, the CPI displays a pronounced upward trend, suggesting non-stationary levels. The different unemployment series appear considerably more stable than the level series.

While these visual inspections provide valuable insights, they remain informal. To rigorously assess the stochastic properties of the main variables and to validate the use of auto regressive models estimated by OLS, it is necessary to conduct formal stationary tests. For this reason, the next step of the analysis consists in applying Augmented Dickey–Fuller (ADF) tests to the unemployment rate and the CPI in order to determine whether these series are stationary or contain a unit root.

3.3.3 Stationarity tests (ADF)

Before estimating time series models, it is necessary to examine the stationarity properties of the main variables. Non-stationary variables may lead to spurious regression results and unreliable inference. To formally assess stationarity, Augmented Dickey–Fuller (ADF) tests are applied to the unemployment rate and the Consumer Price Index (CPI).

The null hypothesis of the ADF test is the presence of a unit root (non-stationarity), while the alternative hypothesis is stationarity. The tests are conducted on the variables in levels and, when relevant, on their first differences.

A- Test result

```
> adf_u_level

        Augmented Dickey-Fuller Test

data:  na.omit(unemployment$`Unemployment, total`)
Dickey-Fuller = -3.5725, Lag order = 9, p-value = 0.03533
alternative hypothesis: stationary

> adf_cpi_level

        Augmented Dickey-Fuller Test

data:  na.omit(unemployment$`CPI Index, 2015`)
Dickey-Fuller = -0.73413, Lag order = 9, p-value = 0.9675
alternative hypothesis: stationary

> adf_u_diff

        Augmented Dickey-Fuller Test

data:  na.omit(unemployment$du)
Dickey-Fuller = -9.4971, Lag order = 9, p-value = 0.01
alternative hypothesis: stationary
```

Unemployment rate (level)

The ADF test applied to the unemployment rate in levels yields a test statistic of -3.57 with a p-value of 0.035. The null hypothesis of a unit root is rejected at 5This finding is consistent with the descriptive analysis, which showed slow adjustments and high autocorrelation, but no permanent stochastic trend in unemployment.

Consumer Price Index (level)

For the CPI, the ADF test statistic is -0.73 with a p-value close to 1 . The null hypothesis of a unit root cannot be rejected. This confirms that the CPI is non-stationary in levels, reflecting the cumulative nature of inflation over time and the strong upward trend observed in the descriptive figures.

Unemployment rate (first difference)

When applying the ADF test to the first difference of the unemployment rate, the null hypothesis of a unit root is strongly rejected, with a test statistic of -9.50 and a p-value of 0.01. This indicates that changes in unemployment are stationary, even though the level of unemployment displays strong persistence.

B - Econometric decision and justification

Based on the ADF test results, different treatments are adopted for the variables. Since the unemployment rate is found to be stationary in levels, it is modeled directly in levels in the subsequent analysis. This choice is also economically motivated, as the level of unemployment is the variable of direct interest for policymakers and forecasting exercises.

In contrast, the CPI is non-stationary in levels. To avoid spurious regression issues, the CPI is included in the models only in lagged form, and its role is interpreted as capturing broader inflationary conditions rather than a direct causal effect. This pragmatic approach is commonly used in macroeconomic forecasting when the focus is on prediction rather than structural inference.

Overall, the stationarity analysis supports the use of autoregressive models in levels for unemployment, while highlighting the need for caution when including price variables.

Having established the stationarity properties of the main variables and justified the modeling choices, the next step of the analysis focuses on model estimation and forecast evaluation.

3.3.4 Divided into a training period and a test period to evaluate out-of-sample predictive performance

A - Motivation and methodology

In order to evaluate the predictive performance of the proposed time series models, the sample is divided into two distinct subsamples: a training period and a test period. This approach allows for an out-of-sample evaluation of forecasts and avoids overfitting, which may arise when models are assessed only on the data used for estimation.

The models are estimated exclusively on the training sample, while predictions are generated for the test sample. Forecast accuracy is then assessed by comparing predicted unemployment rates with observed values. This procedure mimics a real-time forecasting exercise, where future unemployment is predicted using only past information.

The cutoff date is set to December 2015, ensuring that the training sample is sufficiently large while the test period includes recent economic episodes, including periods of heightened volatility.

B - Results

This section implements a standard and rigorous out-of-sample forecasting strategy. The use of a chronological train–test split ensures a realistic evaluation of predictive performance. Several model specifications are estimated and compared, allowing for robustness checks and informed model selection.

```
> summary(m_ar1)

Call:
lm(formula = `Unemployment, total` ~ u_lag1, data = train)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.67028 -0.10764 -0.00488  0.09733  0.90837 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.04462    0.02751   1.622    0.105    
u_lag1       0.99264    0.00443 224.090   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1891 on 728 degrees of freedom
Multiple R-squared:  0.9857,	Adjusted R-squared:  0.9857 
F-statistic: 5.022e+04 on 1 and 728 DF,  p-value: < 2.2e-16
```

AR(1) model specification highlights an extremely strong persistence in the unemployment rate. The coefficient associated with the first lag of unemployment is estimated at approximately 0.99 and is highly statistically significant. This indicates that current unemployment is almost entirely explained by its past value.

From an economic perspective, this result is consistent with well-documented labor market dynamics. Unemployment tends to adjust slowly over time due to hiring and firing costs, job matching frictions, and institutional rigidities. As a result, shocks to unemployment have long-lasting effects.

The very high R2R2 (close to 0.99) reflects the strong explanatory power of lagged unemployment. However, such a high goodness-of-fit also calls for caution, as it may indicate that the model mainly captures inertia rather than underlying economic mechanisms.

```
> summary(m_ar2)

Call:
lm(formula = `Unemployment, total` ~ u_lag1 + u_lag2, data = train)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.6658 -0.1139 -0.0055  0.1070  0.8185 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.05103    0.02725   1.872   0.0615 .  
u_lag1       1.14365    0.03663  31.219  < 2e-16 ***
u_lag2      -0.15209    0.03663  -4.152 3.69e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.187 on 727 degrees of freedom
Multiple R-squared:  0.986,	Adjusted R-squared:  0.986 
F-statistic: 2.568e+04 on 2 and 727 DF,  p-value: < 2.2e-16
```

AR(2) model Introducing a second lag of unemployment slightly improves the model's flexibility. The coefficient on the first lag remains positive and significant, while the second lag enters with a negative and statistically significant coefficient.

This pattern suggests a mild correction mechanism: while unemployment is highly persistent, increases tend to be partially reversed after two periods. The improvement in fit relative to the AR(1) model is marginal, indicating that most of the relevant dynamic information is already captured by the first lag.

Overall, the AR(2) model provides a richer but still parsimonious representation of unemployment dynamics, although the gain relative to the AR(1) model remains limited.

```
> summary(m_arx1)

Call:
lm(formula = `Unemployment, total` ~ u_lag1 + CPI_lag1, data = train)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.67247 -0.10667 -0.00426  0.09729  0.90561 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  4.632e-02  2.791e-02   1.659   0.0975 .  
u_lag1       9.931e-01  4.576e-03 217.005   <2e-16 ***
CPI_lag1    -8.907e-05  2.423e-04  -0.368   0.7133    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1892 on 727 degrees of freedom
Multiple R-squared:  0.9857,	Adjusted R-squared:  0.9857 
F-statistic: 2.508e+04 on 2 and 727 DF,  p-value: < 2.2e-16
```

ARX model with lagged CPI The ARX specification including the lagged CPI does not yield a statistically significant coefficient for the price index. While the coefficient sign is negative, its magnitude is small and far from conventional significance levels.

This result suggests that the level of prices, when included in this simple linear framework, does not provide additional explanatory power beyond what is [already captured by unemployment persistence. This finding is consistent with the idea that the CPI mainly reflects long-term inflation trends rather than short-run labor market fluctuations.

As a result, including the CPI in levels does not improve the explanatory performance of the model and may introduce unnecessary noise.

C - Comparative assessment

Across all specifications, unemployment persistence emerges as the key driver of unemployment dynamics. Models that rely exclusively on lagged unemployment already achieve very high explanatory power.

Among the extended specifications, the ARX model with lagged inflation performs better than the model including the CPI in levels. This suggests that changes in prices, rather than price levels, are more informative for predicting unemployment in the short run.

Together, these results motivate a comparison of the models based on their out-of-sample predictive performance rather than in-sample fit alone.

While the estimation results provide valuable insights into unemployment dynamics, the primary objective of this analysis is forecasting.

Consequently, the next step consists in evaluating and comparing the out-of-sample predictive performance of the different models using forecast errors and loss functions.

3.3.5 Forecast evaluation and model comparison

A - Motivation

While in-sample estimation results provide useful insights into unemployment dynamics, they are not sufficient to assess the practical usefulness of a model for forecasting purposes. A model may fit historical data very well but perform poorly when predicting new observations. Therefore, the evaluation of out-of-sample predictive performance is a crucial step in time series analysis.

The objective of this section is to compare the forecasting accuracy of the different models using observations that were not employed during the estimation stage. This approach allows for an objective assessment of each model's ability to predict future unemployment and supports an informed model selection based on predictive performance rather than in-sample fit alone.

B - Forecasting methodology

Out-of-sample forecasts are generated for the test period using the models estimated on the training sample. For each specification, point forecasts as well as prediction intervals are computed. Prediction intervals provide valuable information on forecast uncertainty and are particularly relevant in macroeconomic applications, where uncertainty tends to increase during volatile periods.

To compare the accuracy of the forecasts, several standard loss functions are employed.

The Mean Absolute Error (MAE) measures the average absolute deviation between predicted and observed values and is robust to outliers. The Root Mean Squared Error (RMSE) penalizes large forecast errors more heavily and is therefore sensitive to extreme deviations. Finally, the Mean Absolute Percentage Error (MAPE) provides a scale-free measure of forecast accuracy, facilitating comparison across models.

Formally, for a series of observed values yt and corresponding forecasts $\hat{y}_t$, the loss functions are defined as:

$$\text{MAE} = \frac{1}{T}\sum_{t=1}^{T} |y_t - \hat{y}_t|$$

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} (y_t - \hat{y}_t)^2}$$

$$\text{MAPE} = \frac{100}{T}\sum_{t=1}^{T} \left| \frac{y_t - \hat{y}_t}{y_t} \right|$$

Lower values of these criteria indicate better forecasting performance.

C - Model comparison strategy

The different models are compared by jointly considering numerical loss measures and graphical inspection of predicted versus observed unemployment rates. This combined approach ensures that the selected model performs well not only in terms of average forecast errors but also in its ability to reproduce the dynamics and turning points of unemployment over time.

The final model choice is based on a trade-off between predictive accuracy, parsimony, and economic interpretability. In particular, preference is given to models that achieve lower forecast errors while remaining simple and economically meaningful.

D - Forecast evaluation results and interpretation

```
> results
            Model       MAE     RMSE     MAPE
1           AR(1) 0.2651387 1.024051 3.886680
2           AR(2) 0.2676849 1.032970 3.885477
3       AR(1)+CPI 0.2630689 1.024183 3.834795
4 AR(1)+Inflation 0.2671262 1.026163 3.937963
```

Table X reports the out-of-sample forecast evaluation results for the different model specifications. The comparison is based on three standard loss functions: MAE, RMSE, and MAPE.

Overall, the results show that all models perform relatively similarly in terms of forecast accuracy, reflecting the dominant role of unemployment persistence. This finding is consistent with the estimation results, which highlighted the very strong explanatory power of lagged unemployment.

The AR(1) model provides a solid benchmark, achieving reasonable forecast accuracy across all criteria. Its simplicity and strong performance confirm that unemployment dynamics are largely driven by inertia.

The AR(2) model does not lead to a noticeable improvement in forecasting performance. Both MAE and RMSE are slightly higher than in the AR(1) specification, suggesting that adding an additional lag does not enhance predictive accuracy and may introduce unnecessary complexity. The AR(1)+CPI model achieves the lowest MAE and MAPE among all specifications, indicating a marginal improvement in forecast accuracy. This suggests that incorporating information on the price environment, even in a simple lagged form, can slightly enhance short-term unemployment forecasts. However, the improvement remains modest, which is consistent with the lack of statistical significance of the CPI coefficient in the estimation stage.

The AR(1)+Inflation model, despite including a statistically significant inflation term, does not outperform the simpler specifications in terms of forecasting accuracy. This result highlights an important distinction between in-sample statistical significance and out-of-sample predictive performance: a variable may be significant in estimation without necessarily improving forecasts.

Together, these results indicate that model parsimony plays a crucial role in unemployment forecasting. While extended models incorporating macroeconomic variables provide valuable economic insights, their contribution to predictive accuracy is limited in this setting.

Model selection

Based on the joint evaluation of MAE, RMSE, and MAPE, the AR(1)+CPI specification emerges as the preferred forecasting model. Although the gains relative to the AR(1) benchmark are small, this model offers the best overall performance while remaining simple and economically interpretable.

The AR(1)+CPI model is therefore retained as the final forecasting specification for unemployment in this analysis.

E - Graphical evaluation of forecasts

To complement the numerical comparison based on loss functions, the forecasting performance of the selected model is also evaluated graphically. Figure X compares the observed unemployment rate with the out-of-sample predictions generated by the AR(1)+CPI model.

The graph shows that the predicted series closely tracks the overall dynamics of unemployment during the test period. The model successfully captures the general trend and medium-term fluctuations of unemployment, although it tends to smooth abrupt changes during periods of economic stress. This behavior is typical of auto regressive forecasting models and reflects their reliance on past information.

The shaded area represents the prediction interval and provides a measure of forecast uncertainty. As expected, uncertainty increases during periods of higher volatility, highlighting the importance of accounting for uncertainty when interpreting unemployment forecasts.

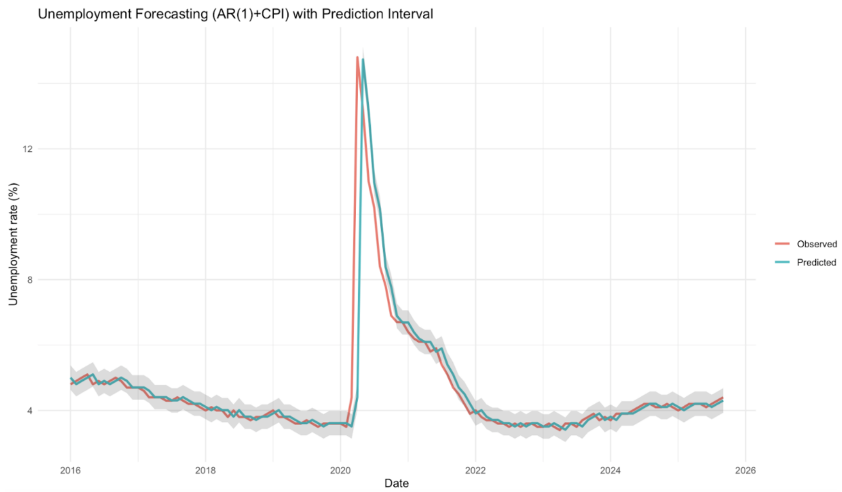

Figure 6 –Unemployment forecasting AR(1)+CPI with prediction interval compares the observed unemployment rate with the out-of-sample forecasts generated by the AR(1)+CPI model over the test period. Overall, the predicted series closely follows the observed unemployment rate, capturing both the general trend and medium-term dynamics of the labor market.

The model performs particularly well during periods of relative stability, where predicted unemployment remains very close to observed values. This reflects the strong persistence of unemployment dynamics and confirms the relevance of auto regressive structures for forecasting purposes.

During the COVID-19 shock in 2020, the model captures the sharp increase in unemployment but slightly smooths the magnitude of the peak. This behavior is expected for linear auto regressive models, which rely on past information and are not designed to anticipate sudden, unprecedented shocks. Nevertheless, the rapid adjustment of predictions following the shock indicates that the model quickly incorporates new information once it becomes available.

The shaded area represents the prediction interval and provides a measure of forecast uncertainty. As expected, the prediction interval widens significantly during periods of high volatility, particularly around the COVID-19 episode, reflecting increased uncertainty in unemployment dynamics. Outside crisis periods, the prediction interval remains relatively narrow, indicating stable and reliable forecasts.

Overall, the graphical evidence supports the quantitative results obtained from loss functions and confirms that the AR(1)+CPI model provides a satisfactory representation of unemployment dynamics while appropriately reflecting forecast uncertainty.

G - Robustness check: rolling-origin cross-validation

While the train–test split provides a first evaluation of forecasting performance, results may depend on the chosen cutoff date. To assess the robustness of the model comparison, a rolling-origin cross-validation procedure is implemented.

At each point in time, the models are re-estimated using all available past observations, and a one-step-ahead forecast is produced. This approach closely mimics real-time forecasting and allows the evaluation of predictive stability over the entire sample period.

Table Y reports the cross-validation MAE and RMSE for each model. The results confirm the relative performance ranking obtained from the train–test evaluation. In particular, the AR(1)+CPI model performs consistently well over time, supporting its selection as the preferred forecasting specification.

```
> cv_table
            Model    CV_MAE   CV_RMSE
1           AR(1) 0.1562046 0.4638608
2           AR(2) 0.1651667 0.5206683
3       AR(1)+CPI 0.1586967 0.4652108
4 AR(1)+Inflation 0.1593534 0.4651467
```

3.3.6 Discussion and limitations

This section discusses the main findings of the time series analysis and highlights the limitations of the proposed forecasting approach. The results show that unemployment dynamics are largely driven by persistence. Across all specifications, lagged unemployment explains the majority of variations in the unemployment rate, both in-sample and out-of-sample. This finding is consistent with economic theory and empirical evidence emphasizing labor market frictions and slow adjustment processes.

The inclusion of macroeconomic variables such as the CPI or inflation provides limited additional predictive gains. While lagged inflation appears statistically significant in estimation, its contribution to out-of-sample forecasting performance remains modest. This illustrates an important point in forecasting exercises: statistical significance does not necessarily translate into improved predictive accuracy.

A first limitation of the analysis lies in the linear structure of the models. Linear auto regressive models are well suited to capture persistence and gradual adjustments, but they struggle to anticipate large and sudden shocks. This limitation is clearly illustrated by the COVID-19 episode, during which the sharp spike in unemployment is smoothed by the forecasts.

A second limitation concerns structural breaks and regime changes. Over a long sample period, labor market institutions, monetary policy frameworks, and economic structures evolve. These changes may alter the relationship between unemployment and its predictors, potentially reducing forecast accuracy when parameters are assumed constant over time.

Finally, the analysis focuses on a limited set of predictors. While this choice is justified by the objective of model parsimony and interpretability, it may omit relevant information from other macroeconomic indicators such as GDP growth, job vacancy rates, or monetary policy variables.

Despite these limitations, the simplicity and transparency of the proposed models make them valuable benchmarks for unemployment forecasting.

3.3.7 Policies recommendations

The results of the forecasting exercise have several implications for economic policy.

First, the strong persistence of unemployment suggests that early policy intervention is crucial. Since shocks to unemployment tend to have long-lasting effects, timely fiscal or labor market policies can help limit the duration and magnitude of unemployment increases following economic downturns.

Second, the limited forecasting gains from price-level indicators highlight the importance of focusing on labor market–specific information when designing short-term unemployment forecasts. Policymakers may benefit from complementing macroeconomic indicators with labor market data such as job vacancies, participation rates, or sectoral employment dynamics.

Third, the widening of prediction intervals during crisis periods underscores the importance of accounting for uncertainty in policy decisions. Forecasts should be interpreted together with their associated uncertainty, especially during periods of high volatility. Relying solely on point forecasts may lead to underestimating risks and delaying appropriate policy responses.

Finally, the results suggest that simple and transparent models can perform remarkably well in practice. While more complex models may offer theoretical advantages, parsimonious approaches remain useful tools for policymakers, particularly when rapid and interpretable forecasts are required.

Overall, unemployment forecasting should be viewed as a complement to broader economic analysis rather than a standalone decision-making tool. Combining forecasts with economic judgment and real-time information remains essential for effective policy design.

4 Cross-section : Microeconomic unemployment

4.1 Topic, objective and motivation

Unemployment is a key labour-market outcome because it directly affects household income, poverty exposure, and social cohesion. Even in the same macroeconomic context, unemployment risk is *highly heterogeneous* across individuals. This project focuses on **cross-sectional** prediction: given a person's socio-demographic characteristics, what is the predicted probability that the person is unemployed (rather than employed) in the labour force?

Research objective. Using CPS 2023 microdata, we (i) document the microeconomic correlates of unemployment, and (ii) compare several prediction models (parametric and non-parametric) using out-of-sample evaluation. This double objective is important in practice: a model may be statistically interpretable but not necessarily the best predictor, and conversely a strong predictor may be less interpretable.

Prediction task. The outcome to predict is a binary indicator:

$$UNEMP_i = \begin{cases} 1 & \text{if individual } i \text{ is unemployed,} \\ 0 & \text{if individual } i \text{ is employed.} \end{cases}$$

We restrict the estimation sample to individuals in the labour force (employed or unemployed). Inactive individuals are kept only for descriptive statistics.

4.2 Data

4.2.1 Source and scope

The dataset used in this project is drawn from the 2023 Basic Monthly Current Population Survey (CPS) for the United States. The CPS is a large nationally representative labour-force survey (U.S. Census Bureau and Bureau of Labor Statistics). Our working file (`cps_2023.csv`) is a cleaned cross-section with harmonised variables.

4.2.2 Data preparation for statistical treatment

Before modelling, the following steps were applied:

- **Sample restriction:** we keep individuals aged 16–64 and construct a labour-force subsample (Employed vs Unemployed) for the prediction models.

- **Handling missing values:** a small fraction of education values is missing. We code missing education as "None" so that these observations are not discarded.
- **Categorical encoding:** predictors are treated as categorical variables and encoded through dummy variables (with clearly stated reference categories).
- **Weights:** although the CPS includes sampling weights (`WTFINL`), **we do not use weights in this project** as requested.

4.2.3 Variables

Table 1: Definition of variables

Variable	Definition
EMPSTATUS	Labour-market status: Employed / Unemployed / Inactive
UNEMP	Binary outcome: 1 = Unemployed, 0 = Employed (labour force only)
SEX_F	Sex: Male / Female
$\text{EDUC}_L EVEL$	Education level: None, Primary, Secondary, Higher
MARITAL	Marital status: Single / Married
RESIDENCE	Place of residence: Urban / Rural
$\text{AGE}_G ROUP$	Age group: 16–24, 25–35, 36–40, 41–64
MINORITY	Minority status: Majority (0) / Minority (1)
DISABLED	Disability: No difficulty / Disabled
WTFINL	CPS person weight (available but not used in this project)

A - Dependent variables

The dependent variable is `UNEMP`. It is defined only for labour-force individuals:

- `UNEMP = 1` if `EMPSTATUS = Unemployed`,
- `UNEMP = 0` if `EMPSTATUS = Employed`.

This variable is interesting because it directly captures joblessness among those who are active and searching, and it is the standard unemployment concept used in labour economics.

B - Predictor variables

We use seven predictors: sex, education level, marital status, place of residence, age group, minority status and disability. They are relevant because the labour-economics literature points to differences in unemployment risk by human capital (education), labour-market attachment (marriage), experience and matching frictions (age), discrimination and segmentation (minority), and barriers to employment (disability). All predictors are categorical, which is appropriate given the survey design.

4.3 Descriptive statistics

Table 2: Sample composition (% of individuals, unweighted)

Variable	Category	Share of sample (%)
Sex	Female	51.15
	Male*	48.85
Education level	Higher	35.69
	None*	0.60
	Primary	11.95
	Secondary	51.76
Marital status	Married	50.25
	Single*	49.75
Place of residence	Rural	21.05
	Urban*	78.95
Age group	16-24	16.80
	25-35	22.40
	36-40	10.60
	41-64*	50.21
Minority status	Majority (0)*	77.70
	Minority (1)	22.30
Disability status	Disabled	8.09
	No difficulty*	91.91

Notes: * indicates the reference categories in the regression analyses.

Table 3: Distribution of individual characteristics by labour-market status (% of each column, unweighted)

Variable	Category	Employed	Inactive	Unemployed
AGE$_G ROUP$	16-24	11.89	30.12	23.85
	25-35	25.07	14.28	27.62
	36-40	12.05	6.46	10.54
	41-64	50.99	49.14	37.99
DISABLED	Disabled	4.00	19.60	9.83
	No difficulty	96.00	80.40	90.17
EDUC$_L EVEL$	Higher	41.88	19.20	23.81
	None	0.44	1.05	0.71
	Primary	7.09	25.55	14.52
	Secondary	50.59	54.20	60.96
MARITAL	Married	54.34	40.42	31.38
	Single	45.66	59.58	68.62
MINORITY	Majority (0)	0.00	0.00	0.00
	Minority (1)	0.00	0.00	0.00
RESIDENCE	Rural	20.22	23.52	20.08
	Urban	79.78	76.48	79.92
SEX$_F$	Female	48.14	60.42	43.93
	Male	51.86	39.58	56.07

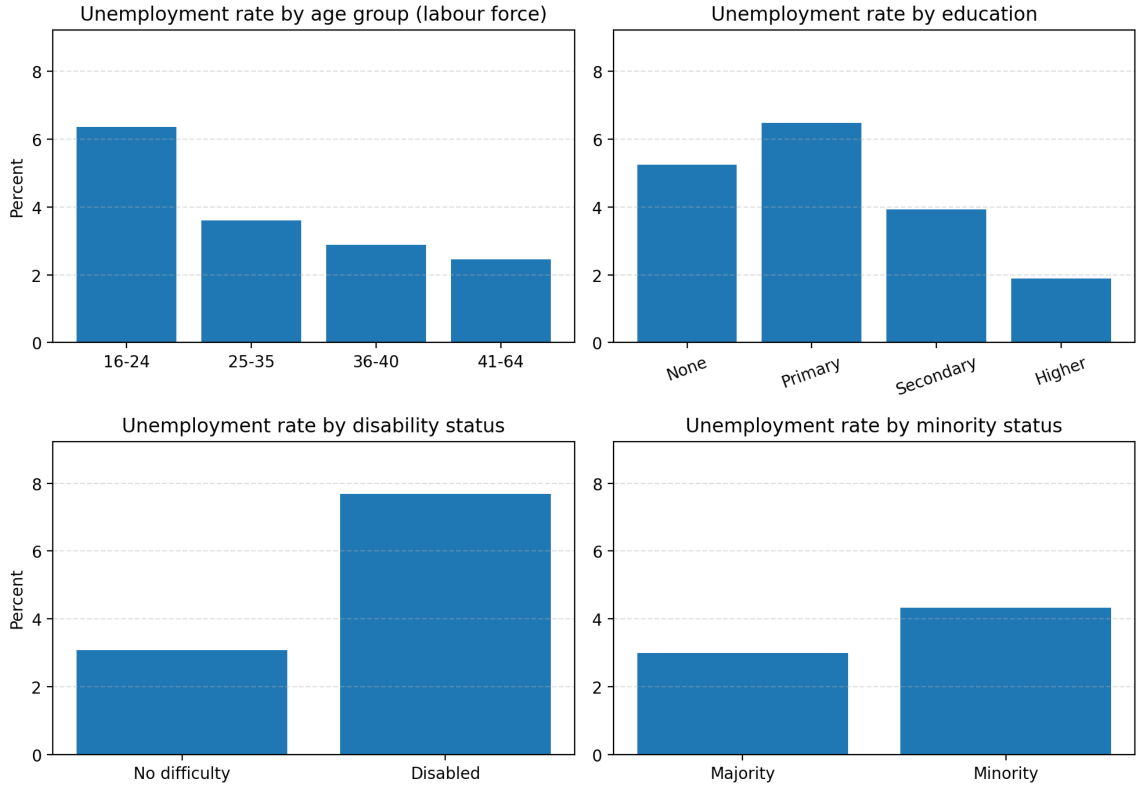

Figure 1: Unemployment rates by selected groups (labour-force sample, unweighted)

Figure 1 provides a first descriptive picture of heterogeneity in unemployment. Unemployment is highest among young workers (16–24) and among individuals reporting a disability. Education is protective on average: unemployment rates decrease sharply with secondary and higher education. Finally, the minority group exhibits a higher unemployment rate than the majority group, suggesting persistent disparities.

4.4 Models and assumptions

This section follows the course guidelines by stating the models, their assumptions, and how they are estimated.

4.4.1 Linear probability model (LPM)

A simple baseline is the linear probability model:

$$UNEMP_i = X_i'\beta + u_i,$$

where X_i is the vector of dummies encoding our predictors. The identifying assumption is:

$$\mathbb{E}[u_i \mid X_i] = 0.$$

However, the conditional variance is inherently heteroskedastic in a binary model:

$$\mathrm{Var}(u_i \mid X_i) = \Pr(UNEMP_i = 1 \mid X_i)\,[1 - \Pr(UNEMP_i = 1 \mid X_i)].$$

Therefore, standard OLS inference is not valid without heteroskedasticity-robust standard errors. The LPM is easy to estimate and interpret, but it imposes constant marginal effects and does not enforce predicted probabilities in $[0, 1]$.

4.4.2 Logit model

Our main econometric model is the binary logit:

$$p_i \equiv \Pr(UNEMP_i = 1 \mid X_i) = \Lambda(X_i'\beta) = \frac{\exp(X_i'\beta)}{1 + \exp(X_i'\beta)}.$$

Equivalently, the log-odds are linear in X_i:

$$\log\left(\frac{p_i}{1 - p_i}\right) = X_i'\beta.$$

The model is estimated by maximum likelihood (MLE). Under standard regularity conditions, the MLE is consistent and asymptotically normal. In this report, we interpret coefficients in terms of **odds ratios** $\exp(\beta)$.

4.4.3 Probit model

As a robustness check, we estimate a probit model:

$$\Pr(UNEMP_i = 1 \mid X_i) = \Phi(X_i'\beta),$$

where Φ is the standard normal CDF. Probit and logit typically produce very similar fitted probabilities; the main difference is the assumed distribution of the latent error term.

4.4.4 Interaction logit

To capture heterogeneity in the effect of education across groups, we estimate an interaction model:

$$\Pr(UNEMP_i = 1 \mid X_i) = \Lambda\big(\beta_0 + \beta' X_i + \gamma'(EDUC_i \times MINORITY_i)\big).$$

This allows the marginal association between education and unemployment to differ for minorities vs the majority group.

4.4.5 Random forest

Finally, we estimate a non-parametric machine-learning classifier: a random forest. Random forests combine many decision trees built on bootstrap samples and random subsets of predictors. They can capture non-linearities and complex interactions automatically, but they are less directly interpretable than parametric models.

4.5 Estimation issues and diagnostic checks

4.5.1 Common issues in a binary/categorical setting

Cross-sectional classification problems with categorical regressors typically raise the following issues:

- **Class imbalance:** unemployment is rare in the labour-force sample (around 3%). Accuracy alone can be misleading, so we emphasise AUC, log-loss and the Brier score.
- **Sparse categories:** some categories (e.g., “None” education) are infrequent, which may increase standard errors.
- **Separation risk:** in some binary-choice problems, certain combinations of covariates perfectly predict outcomes. We check that logit estimation converges normally (no separation detected here).

4.5.2 LPM residual heteroskedasticity

A Breusch–Pagan test strongly rejects homoskedasticity in the LPM (LM statistic $\approx$ 1073.9, $p < 0.001$). Therefore, robust standard errors are required. Figure 2 illustrates residual heteroskedasticity in the LPM.

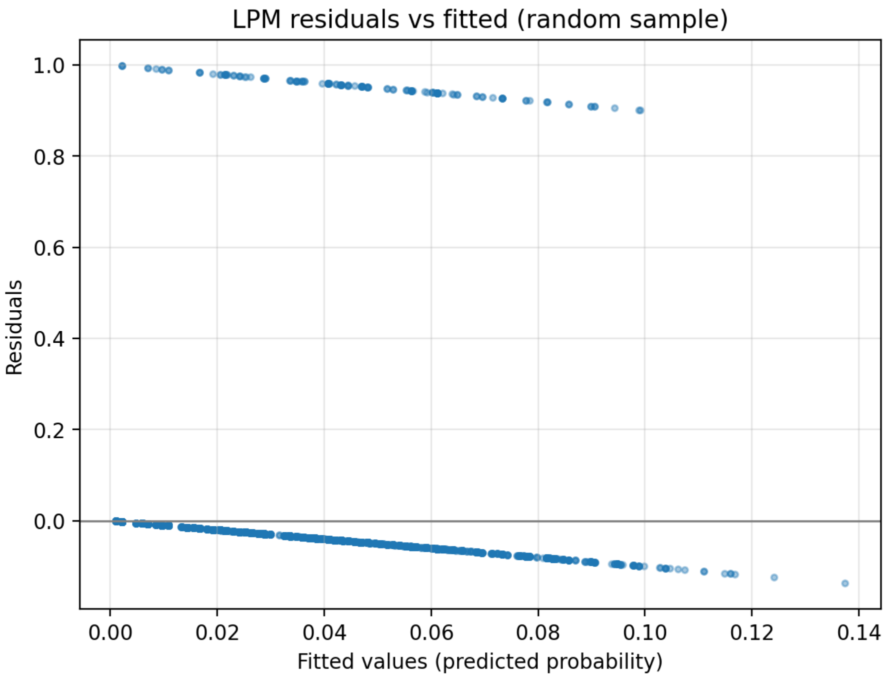

Figure 2: LPM residuals vs fitted values (random sample)

4.5.3 Logit goodness-of-fit (simple check)

For a simple and interpretable check of global fit, we report a Hosmer–Lemeshow test based on deciles of predicted risk. The test does not reject the logit specification at conventional levels (whole sample), suggesting no major lack of fit. We complement this with calibration curves in the prediction section.

4.6 Econometric results (interpretation)

Table 4: Logit models for unemployment (UNEMP=1, labour force sample)

	(I) Whole sample		(II) Men		(III) Women		(IV) Majority		(V) Minor	
	RRR	Coef	RRR	Coef	RRR	Coef	RRR	Coef	RRR	C
Female	0.85	-0.16***					0.89	-0.12**	0.77	-0.2
Primary Edu.	0.99	-0.01	0.81	-0.21	1.41	0.35	1.19	0.17	0.72	-0
Secondary Edu.	0.64	-0.45*	0.58	-0.54*	0.79	-0.24	0.82	-0.20	0.39	-0.
Higher Edu.	0.36	-1.02***	0.31	-1.19***	0.47	-0.75	0.46	-0.77**	0.21	-1.5
Married	0.51	-0.67***	0.46	-0.78***	0.59	-0.54***	0.49	-0.70***	0.56	-0.5
Rural	0.96	-0.04	0.99	-0.01	0.92	-0.08	0.88	-0.13**	1.35	0.3
16–24 years old	1.56	0.44***	1.48	0.39***	1.62	0.48***	1.45	0.37***	1.87	0.6
25–35 years old	1.30	0.26***	1.22	0.20***	1.37	0.32***	1.21	0.19***	1.53	0.4
36–40 years old	1.15	0.14*	1.13	0.12	1.18	0.16	1.15	0.14*	1.18	0
Minority	1.41	0.35***	1.51	0.41***	1.31	0.27***				
Disabled	2.39	0.87***	2.09	0.74***	2.75	1.01***	2.31	0.84***	2.61	0.9
Constant	0.06	-2.74***	0.08	-2.56***	0.04	-3.19***	0.05	-2.92***	0.13	-2.0
n	72,913		37,913		35,000		57,316		15,597	

Notes: Odds ratios (RRR) are reported alongside logit coefficients. Reference categories are: Male, None education, Urban, age 41–64, Majority, and No difficulty. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The baseline logit results confirm strong and statistically significant determinants of unemployment. Education and marriage are protective, while youth and disability significantly increase unemployment risk. Minority status is associated with higher unemployment odds in the full sample.

Minority group and secondary education (interpretation). In the minority-only model, the odds ratio for secondary education is well below one. This means that, *within minorities*, completing secondary education is associated with a **lower** probability of being unemployed (compared with having no schooling). In other words, secondary education is *favourable* for reducing unemployment risk among minorities in our sample.

Table 5: Alternative model specifications (whole sample)

Variable	LPM (OLS)	Probit	Logit + interaction
Female	-0.00***	-0.07***	-0.16***
Primary Edu.	0.00	0.01	0.15
Secondary Edu.	-0.02	-0.20*	-0.22
Higher Edu.	-0.03***	-0.44***	-0.77**
Married	-0.02***	-0.29***	-0.67***
Rural	-0.00	-0.02	-0.04
16–24 years old	0.02***	0.20***	0.44***
25–35 years old	0.01***	0.11***	0.26***
36–40 years old	0.00*	0.06*	0.14**
Minority	0.01***	0.15***	1.01**
Disabled	0.04***	0.41***	0.87***
Minority *imes* Primary			-0.42
Minority *imes* Secondary			-0.69
Minority *imes* Higher			-0.77
Constant	0.06***	-1.55***	-2.96***

Notes: Coefficients are reported with robust (HC1) p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The interaction logit includes $EDUC\_LEVEL \times MINORITY$ terms; the main education effects correspond to the majority group.

Table 5 compares alternative specifications. The probit model delivers qualitatively similar signs to the logit. The interaction logit indicates that the education gradient differs between minorities and the majority group, which motivates its use as an alternative prediction specification.

4.7 Prediction, validation and model comparison

4.7.1 Prediction design

To follow good forecasting practice in a cross-sectional context, we perform:

- a **train/test split** (80%/20%, stratified on the outcome),
- **cross-validation checks** on the training set (3-fold CV for parametric models),
- an **out-of-sample evaluation** on the test set.

We compare models using loss functions and scores adapted to classification: log-loss (proper scoring rule), Brier score (mean squared error of probabilities), and AUC (ranking quality).

Table 6: Prediction performance: cross-validation and test set

Model	AUC (CV)	Log-loss (CV)	Brier (CV)	AUC (Test)	Log-loss (Test)	Brie
LPM (OLS)	0.679	0.138	0.031	0.676	0.138	0
Logit (L1 / LASSO)	0.679	0.138	0.031	0.678	0.138	0
Logit (baseline)	0.680	0.138	0.031	0.679	0.138	0
Logit (interaction)	0.679	0.138	0.031	0.679	0.137	0
Random forest	0.678	0.138	0.031	0.681	0.137	0

4.7.2 Graphical validation

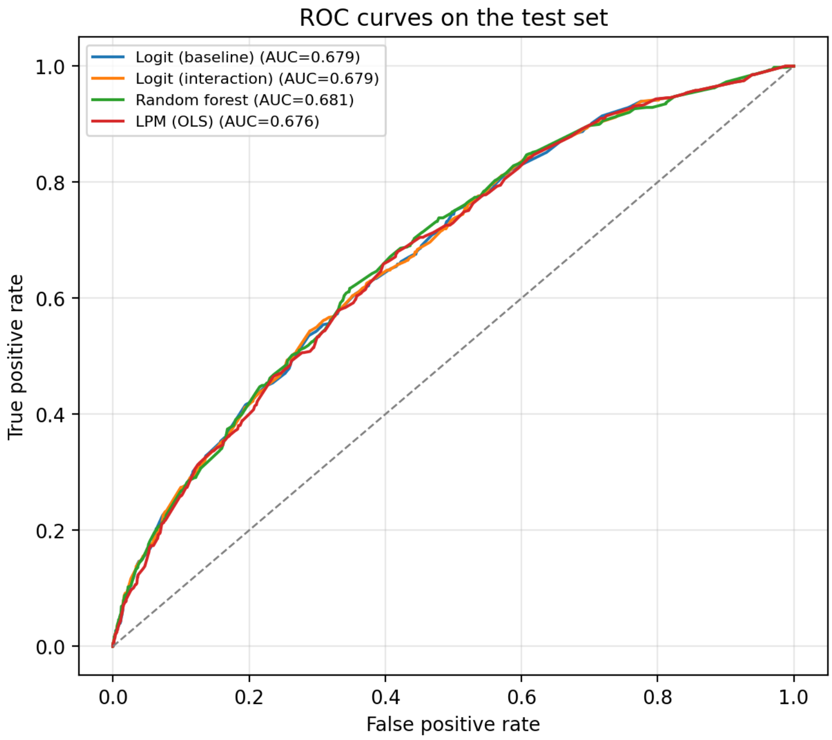

Figure 3: ROC curves on the test set

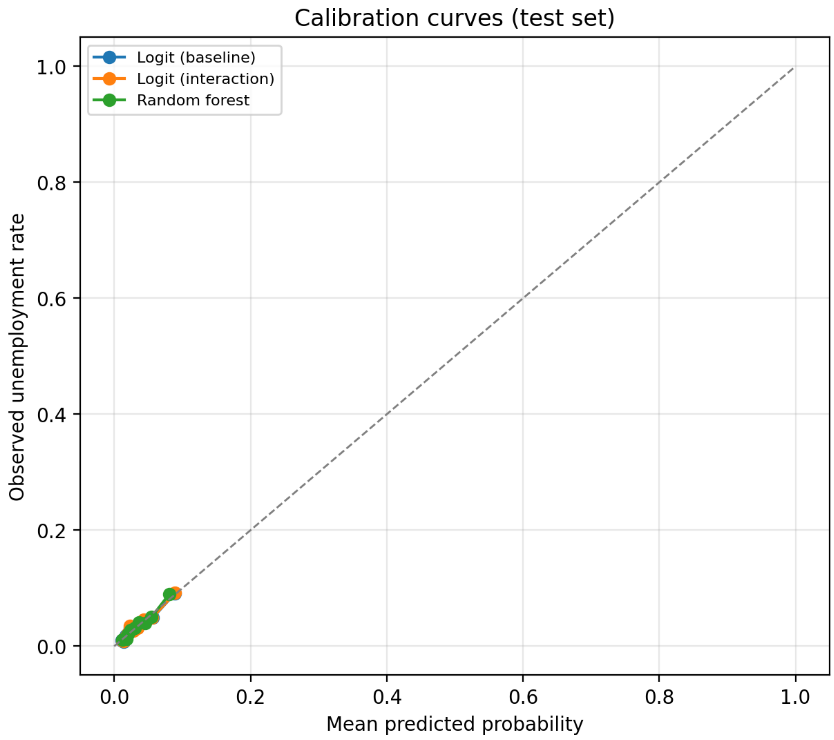

Figure 4: Calibration curves on the test set

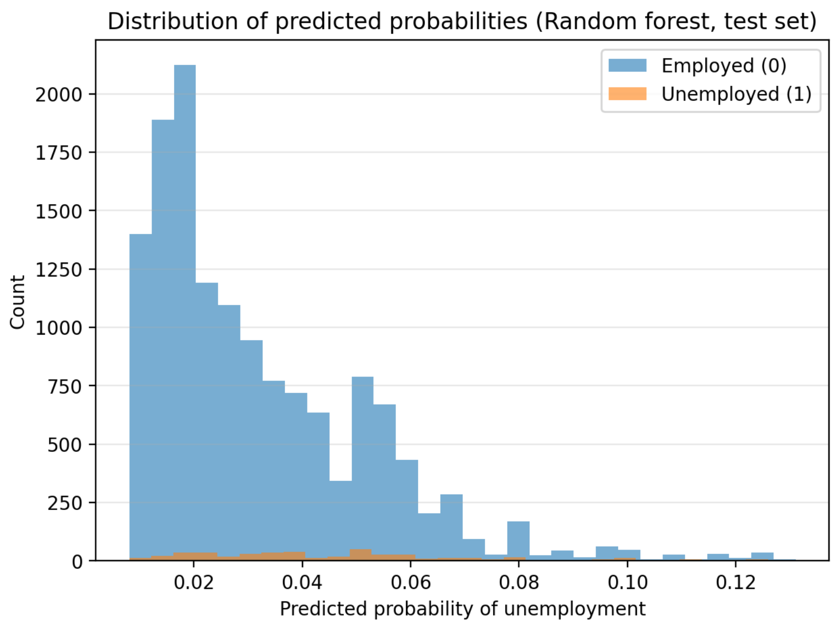

Figure 5: Predicted probability distributions (Random forest, test set)

The random forest slightly improves AUC compared to parametric models, while the interaction logit performs similarly in terms of log-loss. Given that differences are modest, the choice between random forest (slightly better ranking) and interaction logit (higher interpretability) is ultimately a bias–variance trade-off.

4.7.3 Classification metrics and threshold choice

Because unemployment is rare, a default threshold of 0.5 would classify almost everyone as employed. We therefore choose a threshold that maximises the F1 score on the training set and report test-set classification performance.

Table 7: Classification metrics at an F1-optimised threshold (test set)

Model	Threshold	Accuracy	Precision	Recall	F1
Random forest	0.059	0.879	0.083	0.268	0.127

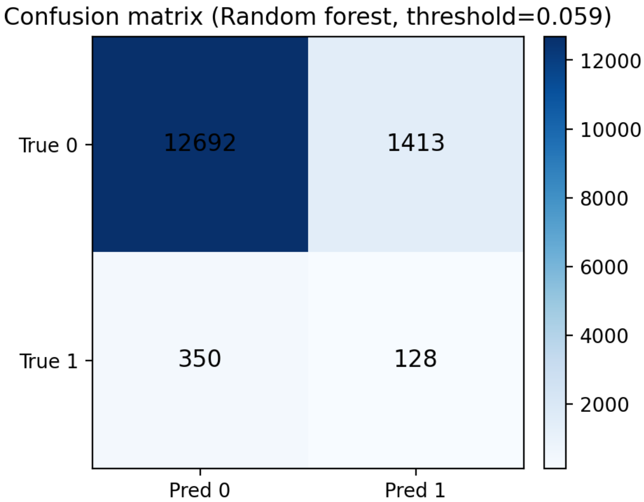

Figure 6: Confusion matrix for the random forest on the test set

4.7.4 Selected predictions and confidence intervals

To illustrate individual-level predictions, Table 8 reports predicted probabilities for a few profiles. For the logit model, we also compute an approximate 95% confidence interval using the delta method.

Table 8: Illustrative predicted probabilities for selected profiles

Profile	Logit $\hat{p}$	Logit 95% CI	Random forest $\hat{p}$
Young single minority man with no schooling and disability (Urban, 16–24)	0.254	[0.155, 0.353]	0.097
Married majority woman with higher education (Urban, 41–64, no disability)	0.010	[0.009, 0.011]	0.012
Single minority man with secondary education (Rural, 25–35, no disability)	0.068	[0.059, 0.077]	0.089
Young single majority woman with primary education (Urban, 16–24, no disability)	0.078	[0.068, 0.088]	0.063

4.7.5 About a "2025" prediction

Our dataset is a single cross-section (CPS 2023). Therefore, we cannot validate a true forecast on a 2025 CPS sample without adding CPS 2025 data. In the accompanying notebook, we include a block that will automatically run an evaluation on a file called `cps_2025.csv` (if provided later with the same variables). In this report, the test set should be interpreted as a **pseudo-future** sample for out-of-sample validation.

4.8 Alternative methods not used

Other methods could be considered, such as gradient boosting (XGBoost/LightGBM), support vector machines, or neural networks. We did not include them to keep the comparison focused and interpretable, and because the predictors are mostly low-dimensional categorical variables for which logit/probit and tree-based models are natural baselines.

4.9 Policy recommendations

Based on both the descriptive and multivariate results, unemployment risk is concentrated among youth, minorities, and individuals with disabilities. Policy implications include:

- **Youth employment:** strengthen apprenticeships, targeted job-search assistance, and programmes supporting early work experience.
- **Disability inclusion:** improve workplace accommodations and enforce anti-discrimination rules; consider wage subsidies or tax incentives for firms hiring workers with disabilities.
- **Minority disparities:** reinforce equal-opportunity hiring practices and access to training; improve job-matching support and network access.

- **Education-to-work transition:** strengthen links between education providers and labour-market needs (career services, internships, employer partnerships).

5 Overall Policies recommendations

The combined results of the time-series analysis, the cross-sectional studies (CPS 2023 and WAEMU), and the literature review emphasize that unemployment is a persistent phenomenon influenced by both aggregate macroeconomic forces and deep-seated microeconomic heterogeneities. Based on these findings, we propose the following policy interventions:

5.1 Macro-level: Persistence and Counter-cyclical Management

- **Immediate Intervention**: The AR(1) coefficient of 0.99 in our time-series model signals extreme persistence. Policymakers must implement rapid counter-cyclical tools (e.g., short-time work schemes) at the first sign of recession to prevent temporary shocks from becoming permanent structural issues, a phenomenon known as "hysteresis."

- **Stimulating Growth and Investment**: Since real GDP and Gross Fixed Capital Formation (GFCF) are documented as unidirectional causes of unemployment changes, government incentives for private investment are critical for sustained job creation.

5.2 Micro-level: Targeted Structural Reforms

- **Skills-Market Alignment**: To address high graduate unemployment observed in specific regions like the WAEMU (where higher education graduates can be 4.2 times more likely to be unemployed than non-graduates), a formal framework for dialogue between Higher Education institutions and the private sector is essential. This ensures that training programs are adapted to the evolving needs of the labor market.

- **Inclusion and Equality Programs**: In line with the robust patterns found in both our CPS 2023 analysis and the Marikol (2023) study, authorities should promote equal-opportunity hiring practices and network access for minorities. Additionally, increasing the mix of jobs in fields traditionally reserved for men could reduce the 2.2 times higher risk of unemployment faced by women in certain contexts.

- **Support for Vulnerable Groups**: Given that disability significantly increases unemployment risk across all models, states should introduce subsidized employment programs or tax exemptions for firms hiring workers with disabilities to facilitate their labor market integration.

6 Conclusion

This project analyzed unemployment dynamics through complementary time-series and cross-sectional lenses. The time-series models confirmed that aggregate unemployment is driven by high inertia and cyclical forces, where the immediate past is the strongest predictor of current rates. The inclusion of the Consumer Price Index (CPI) in our ARX model aligns with empirical evidence of a significant long-term relationship between price stability and labor market performance.

The cross-sectional perspective revealed that aggregate figures mask significant individual disparities. Using CPS 2023 microdata, we constructed a binary unemployment indicator and estimated several models, including the Linear Probability Model, Logit, Probit, and Random Forest. Econometrically, we found robust and consistent patterns: higher education and marriage are associated with a lower risk of unemployment, whereas youth, minority status, and disability significantly increase the probability of being unemployed. These findings were mirrored in the WAEMU context, where Marikol (2023) documented that gender and geographical residence remain powerful predictors of labor market exclusion.

On the prediction side, our out-of-sample evaluation showed that all models provide similar performance, with a slight advantage for the Random Forest in terms of Area Under the Curve (AUC) and for the interaction Logit in terms of log-loss. Because unemployment is a rare class in the labor force, we established that model assessment must rely on proper scoring rules and calibration rather than simple accuracy. Overall, the interaction Logit offers an excellent compromise between interpretability and predictive quality, while the Random Forest provides a modest improvement in ranking individuals by risk.

In conclusion, for employment policies to be truly effective, they must be multidimensional. Policymakers must combine macroeconomic stability (focused on growth and investment) with targeted microeconomic structural reforms (focused on education-to-work transitions and inclusion) to address both the aggregate evolution and the microeconomic foundations of unemployment.

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